Source log

Field	Value
Dataset	IBM Telco Customer Churn (sample data, fictional company)
Actual download source	**Kaggle blastchar/telco-customer-churn**
Download URL	https://www.kaggle.com/api/v1/datasets/download/blastchar/telco-customer-churn
Dataset page	https://www.kaggle.com/datasets/blastchar/telco-customer-churn
Original file	WA_Fn-UseC_-Telco-Customer-Churn.csv
Downloaded (UTC)	2026-10-03T11:36:37.024155+00:00
Version	Snapshot identified by SHA-256; upstream version not supplied by download endpoint; Kaggle metadata reports version 1 (last updated 2018-02-23T18:20:00.21Z)
SHA-256	88be4b93fbe0cc83421af1c503794c97c342eca914c1576db7c276e61d61358a
Rows x columns	7,043 x 21
Licence	Kaggle metadata: "Data files © Original Authors". Dataset attribution: IBM sample / BlastChar. Consult source data card for distribution rights; software license does not extend to source data.
Joins	IBM extended sample joined one-to-one on customerID (7,043 of 7,043 matched; 7 shared fields agree for every customer); only churn reasons and the CLTV index are kept, in a separate file used for description, never as model inputs
Filters	Exact duplicate rows removed from the full file before splitting; no other rows filtered (0 rows removed)

Second source: churn reasons (joined for description only)

Field	Value
Dataset	IBM Telco customer churn, extended version (same 7,043 fictional customers, 33 columns) (TanKY, 2020)
Download source	**Kaggle yeanzc/telco-customer-churn-ibm-dataset**
Download URL	https://www.kaggle.com/api/v1/datasets/download/yeanzc/telco-customer-churn-ibm-dataset
Dataset page	https://www.kaggle.com/datasets/yeanzc/telco-customer-churn-ibm-dataset
Original file	Telco_customer_churn.xlsx
Downloaded (UTC)	2026-10-10T14:48:44.721775+00:00
SHA-256	1bcbc0ccc9b352175216979102628e579cfbde2c3ff57b005de168a433122640
Rows x columns	7,043 x 33
Licence	IBM Cognos Analytics sample data republished on Kaggle; consult the data card for distribution rights.
Join	One-to-one on customerID = CustomerID: 7,043 of 7,043 customers matched; Churn 100%, tenure 100%, MonthlyCharges 100%, Contract 100%, InternetService 100%, PaymentMethod 100%, TechSupport 100% agreement
Fields kept	Churn Reason, CLTV, stored in data/processed/customer_context.csv, separate from the modelling file
Fields not used	Churn Score, location and all other columns. IBM's churn score comes from IBM's own model with undocumented inputs, so it could carry information unavailable before a customer leaves; it is never stored

Churn reasons are recorded only for customers who left, after they left, so they can never be a model input; they are used to judge which offers can plausibly work (analysis report). The CLTV index is IBM's undocumented formula and is used only to cross-check the value proxy.

Provenance notes

- IBM publishes the data as a sample for its analytics products: "a fictional telco company that provided home phone and Internet services to 7043 customers in California in Q3" (IBM, 2019; IBM docs). The Kaggle copy is the 21-column base version (BlastChar, 2018).
- The churn window differs between descriptions: "left within the last month" (Kaggle and the base IBM sample) and "left the company this quarter" (IBM extended sample). The project treats the label as churn within one observation window and does not annualise it.
- Currency is not stated by the source; amounts are treated as USD for a US company.
- The pipeline first tries the Kaggle API, then the IBM GitHub copy (IBM GitHub). The raw file is never modified; on re-runs the checksum is verified and the pipeline stops if it changed.

Synthetic elements

The offer table (offer costs, contact cost, save rates, eligibility rules) is created by the team for simulation, as the brief allows. It is version v2 in configs/default.yaml and is documented with sources in the assumption register. No randomised campaign data exist in the source.

Cleaning log

Step	Decision	Rows affected
Whitespace	Trim leading/trailing spaces in every text field	all
Duplicates	Drop exact duplicate rows from the full file before splitting	0
TotalCharges blanks	Blank strings become missing; customers with tenure 0 get 0 (they have not been billed yet)	11 of 11
Other numeric gaps	Median imputation fitted on train inside the model pipeline; the same train medians fill gaps before valuing customers	0 in the source
SeniorCitizen	Normalised to the strings "0"/"1"	all
Label	Churn Yes/No mapped to churn 1/0; not used as a feature	all
Service consistency	"No phone service" / "No internet service" kept as their own categories and checked against the parent service; as model inputs they count as "does not have the service"	0 violations
Extended-sample join	Join on customer ID only after every shared field agrees; keep churn reason (churners only) and CLTV in a separate file; map the 20 reasons to IBM's five categories	7,043 matched
Outliers	Valid extreme values kept; nothing trimmed	0
Split	Stratified 60/20/20 train/validation/test, seed 42; IDs saved in data/processed/splits.csv	4,225 / 1,409 / 1,409

Quality summary

Metric	Value
Rows before / after	7,043 / 7,043
Unique IDs	7,043
Missing values after cleaning	0

Churn counts (0 / 1)	5,174 / 1,869

Validation rules live in src/telco_retention/data.py (validate). The same rules reject invalid uploaded CSV files in the dashboard, listing the row, column and reason.